D1_Detailed_Design

Assignment 6 - Detailed System Design, Part 1 (Design D1)

1. Header, scope, and conventions

Project title: Pulse - platform-independent IoT communication for resilient farm sensor networks.

Goal statement: Move every farm sensor reading from a field device to a trusted Hub over any available link, and keep each reading until the Hub stores it.

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This document details the Node (C3) and Hub (C5); D2 details the Sensor Driver (C1), Edge (C2), Transport Adapter (C4), and Builder (C6).

The Node owns reliable delivery and sender checks. The Hub owns the data model and the storage transaction. Jacob owns C3. Yatharth owns C5. Sujal checks sensor formats with both owners. The shared Protobuf package defines I3, I5, and I6. It does not run as a component.

Figure conventions

Figure 1 uses blue rectangles for Hub tables. Each rectangle lists the table name and its key fields. Solid lines show foreign-key relationships. Labels give the count at both ends: `1` means exactly one, `0..*` means zero or more, and `1..*` means one or more. PK marks a primary key. FK marks a foreign key. One composite key uses all marked fields together. Lines do not show message flow. The figure uses no person or hardware symbols. The SQL source gives the full constraints. The Hub transaction enforces the minimum of one Reading per Envelope.

Basis and changes from D0

The project title, goal, component IDs, and reading key stay the same as D0 [P1]. The following changes make D0 precise.

Item	D1 decision and basis
Retry limit	Retry every 30 seconds until the 3,600-second deadline. Five retries alone do not cover US-01. This changes D0's retry limit.
Queue key	Use `(message_id, destination)` per field Node. This prevents one destination's receipt from removing another destination's copy.
Hub model	Add Origin, Sensor, and Envelope tables around the existing Reading key. Compare duplicate content before acknowledging it.
Carrier and identity	Current code uses Reticulum over TCP, a 16-byte identity hash, and no Pulse TTL. D0 names UDP and a 32-byte sender key.
Build status	These refinements define target behavior. Current code still uses message-only queue keys, five retries, and a flat readings table [P3].

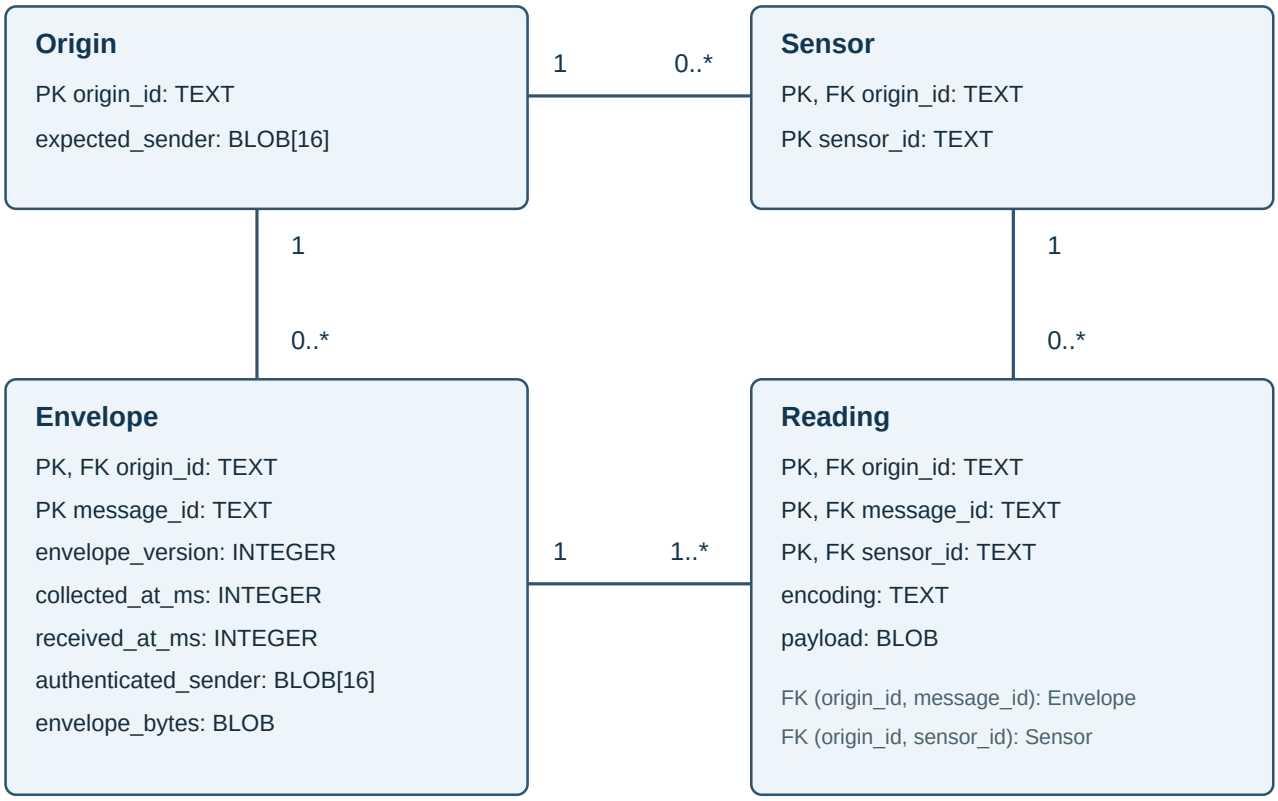
The queue stays in memory in D1. Outage tests keep both Nodes running. D2 must define a durable queue before Pulse claims recovery after a Node restart. A local acceptance response confirms memory admission. It does not confirm storage at the Hub.

2. Data model, the D1 diagram

Figure 1 details C5. All four tables use SQLite on the Hub host. The diagram fits on this page.

Figure 1. C5 Hub: relational data model

Target D1 schema. PK = primary key; FK = foreign key.



A2 enforces one or more Readings per Envelope in one transaction. Lines show relationships, not message flow.

Structural decisions

Decision	Reason and requirement
Envelope is an entity	One publication groups samples into one commit and one receipt. Its stored content identifies conflicting retries. US-01, US-02; I6, I7.
Sensor is an entity	An origin can use one sensor across many messages. An encoding belongs to each Reading because formats can change. US-04.
Use one-to-many links	Each Envelope has one Origin. Each Reading has one Envelope and one Sensor. Reading joins their histories. US-02, US-04.
Use a relational store	Keys, foreign keys, and transactions protect a complete publication. A plain file needs custom checks and recovery. US-02, US-05.

An Origin represents a configured device name. Its expected sender hash binds that name to one identity. The Node trust table remains the authority. The Hub imports the same binding before it accepts remote data. One identity can own several Origin names. Key changes require a controlled configuration update; D1 does not permit two active keys for one Origin.

Key attributes, constraints, and indexes

Table	Key attributes and rules
Origin	`origin_id: TEXT`, PK; `expected_sender: BLOB`, exactly 16 bytes. The name contains 1-64 ASCII characters. Provision this row before delivery.
Sensor	`(origin_id, sensor_id): TEXT`, PK; `origin_id` is an FK. Provision sensors before delivery. Both names contain 1-64 ASCII characters.
Envelope	`(origin_id, message_id): TEXT`, PK; `origin_id` is an FK. Store `envelope_version: INTEGER = 1`, `collected_at_ms: INTEGER >= 0`, `received_at_ms: INTEGER > 0`, `authenticated_sender: BLOB` of 16 bytes, and `envelope_bytes: BLOB`.
Reading	`(origin_id, message_id, sensor_id): TEXT`, PK. The first two fields form an FK to Envelope. `(origin_id, sensor_id)` forms an FK to Sensor. Store `encoding: TEXT` and `payload: BLOB` unchanged.

All fields are required. `collected_at_ms = 0` means the sensor time is unknown. All other times use UTC milliseconds since the Unix epoch. The receiver clock supplies `received_at_ms`. Arrival time does not replace collection time. Payload units belong to the decoder contract. For example, the moisture value is a fraction from 0 to 1.

Index	Lookup and reason
Envelope PK	Find a duplicate by origin and message in logarithmic time. US-02; I7.
Reading PK	Enforce D0's unique sample key and fetch a whole envelope. US-02, AC-01.2.
Sensor PK and Origin PK	Check the configured sensor and identity binding. US-04, US-05.
Envelope `(origin_id, collected_at_ms, message_id)`	Read a device's time range without a full scan. The final field resolves equal times. US-01.
Reading `(origin_id, sensor_id, message_id)`	Fetch sensor history and check references before sensor removal. US-04.

Keep records for a planning period of 30 days. Keep the Envelope and its Readings together for at least 3,600 seconds after receipt. Never delete the duplicate key while a valid retry can arrive. D2 defines removal and backup rules. Use `foreign_keys=ON`, `journal_mode=WAL`, `synchronous=FULL`, and a 5,000 ms busy timeout. FULL adds a disk sync at each commit in WAL mode [W1].

C3 state model

`Publication` holds an immutable envelope and one unique `message_id: string` for the Node. `Delivery` holds one destination's state. One Publication has `1..*` Deliveries; each Delivery has exactly one Publication. The Delivery key is `(message_id, destination)`. Its fields are `expected_peer: bytes[16]`, `ack: AckMode`, `attempts: uint32`, `accepted_at: Instant`, `deadline: Instant`, `next_attempt: Instant`, `wrapper: bytes`, and `notify: set<uint64>`. `Instant` is a monotonic local time and has no wire format.

Use a hash map for Deliveries and one queue for new work. Capacity includes queued and waiting reliable Deliveries. Set the field limit to 128 Deliveries. Keep best-effort traffic in a separate 64-item queue. A full reliable queue rejects new work. A full best-effort queue removes its oldest item and increases `dropped`. Do not add a durable queue in this design.

3. Core algorithms

A1. Retry delivery until the requested receipt or expiry

1. Purpose. A1 keeps reliable publications during link outages. It prevents an early receipt from deleting data before the Hub commit. It implements US-01 and US-02 through I3, I4, I5, and I6.

2. Inputs and outputs. Inputs are `PublishRequest`, verified `StatusUpdate`, `now: Instant`, and configured limits. Section 5 gives exact wire types. Output is `Result<accepted, ErrorDetail>`, zero or more signed packet sends, and `StatusEvent` values. Internal `accepted` is a boolean. Status values are `RECEIVED`, `STORED`, `EXPIRED`, or `FAILED`.

3. Procedure and complexity. Check the request and encode its signed wrapper. Reject a remote wrapper above 383 bytes. Reject a duplicate message ID with different envelope bytes. Admit one Delivery per destination. A repeat of the same pending request reuses the Delivery without extending its deadline. Add the caller to its notification set. Set the first attempt to now and the deadline to acceptance plus 3,600 seconds.

At each one-second tick, check expiry before sending. Expire each Delivery whose deadline is at or before now. For each remaining due Delivery, send its immutable wrapper. Increase attempts and set the next attempt to now plus 30 seconds. A send error keeps the Delivery. A delayed tick sends once; it does not send a burst of missed attempts. The interval is a minimum; the carrier can delay a send.

Verify each remote receipt's signature and expected peer. Resolve the destination from that peer's configured service. For `ACK_STORED`, `RECEIVED` does not remove the Delivery. A valid `STORED` removes only that destination's Delivery. For `ACK_RECEIVED`, a valid `RECEIVED` or `STORED` completes it. A remote `FAILED` or `EXPIRED` does not prove storage. Keep an `ACK_STORED` Delivery until local expiry. Delete the Publication only when it has no Deliveries.

Admission and receipt lookup take expected `O(1)` plus `O(B)` encoding and signature work. A tick takes `O(Q + D*B)`, where Q is pending Deliveries, D is due sends, and B is wrapper bytes. Memory takes `O(Q*B)` plus notification sets. The scan uses library containers; it does not require custom sorting.

Planning size	Estimated work and effect
One field Node	One message per 60 seconds gives at most 60 pending messages in one hour, with one destination. A tick checks about 60 entries.
100 times that load	A shared test Node holds 6,000 Deliveries. A tick checks 6,000 entries. Raise its test capacity to 8,192. This does not fit the field limit.
Storage estimate	At the 383-byte wrapper limit, 60 wrappers use about 23 KB; 6,000 use about 2.3 MB, before container overhead.

4. Why this algorithm. A simple scan suits the bounded field queue. A minimum heap reduces idle checks to `O(log Q)` updates but adds cancellation state. Stop-and-wait blocks other sensors during an outage. Five retries end after about 150 seconds and cannot cover US-01. At 100 times load, measure scheduler time and carrier capacity before selecting a heap. Radio airtime can dominate these CPU costs.

5. Edge cases. An empty queue sends nothing. Equal retry times each get one send. Their order does not affect storage. Repeated receipts after completion do nothing. A wrong peer, bad signature, or unknown message does not remove data. Full capacity returns `QUEUE_FULL` without removal. At exactly the deadline, expiry wins. With several services on one peer, assign a distinct message ID per destination until a future protocol carries destination in each receipt. Node restart loses memory state; D2 must resolve that limit.

A2. Check and commit a complete envelope once

1. Purpose. A2 converts a verified delivery into one stored row per sensor. It sends `STORED` only after a complete commit. It implements US-02, US-04, and US-05; AC-01.1 to AC-01.3; I6 and I7.

2. Inputs and outputs. Input is `DeliveryEvent{destination: string, message: PulseEnvelope, authenticated_sender: bytes[16]}` plus `receivedAtMs: int64 > 0`. Output is `StoreResult = STORED | DUPLICATE` or `StoreError`. `StoreError` is `INVALID_ENVELOPE | UNKNOWN_SENSOR | UNSUPPORTED_ENCODING | INVALID_PAYLOAD | IDENTITY_MISMATCH | CONFLICTING_DUPLICATE | STORAGE_ERROR`. These are internal codes, not new Protobuf enum values. Success causes `AckRequest{message_id, status: STORED, detail: ""}`. Failure causes a log entry and no receipt.

3. Procedure and complexity. Check the envelope fields and sender binding. Check each sensor against the Sensor table. Reject repeated sensor IDs within one envelope. Resolve each encoding in the decoder registry. Use a library parser for structured data. Reject the whole envelope if any sample fails. Do not use a prefix character as a JSON validity check.

Encode the parsed envelope with the pinned Java Protobuf codec. Store this generated representation as `envelope_bytes`. This uses the same codec for every comparison; it does not compare arbitrary Protobuf field order on the network. Start one `BEGIN IMMEDIATE` transaction. Look up `(origin_id, message_id)`. If it exists, compare its stored bytes and sender with the new values. If both match, commit the read decision and return `DUPLICATE`. If either differs, roll back and return `CONFLICTING_DUPLICATE`. A retry must keep sample order and payload bytes unchanged.

If the key is new, insert one Envelope and all its Readings. Commit the transaction. Return `STORED` and send the receipt. A failed commit rolls back all new rows. If the receipt send fails after commit, retain the rows. The next matching retry returns `DUPLICATE` and sends the receipt again. The PK check inside the transaction handles concurrent writers. Never use `INSERT OR IGNORE` to hide a data conflict.

Let S be samples, B be envelope bytes, R be stored Readings, and E be stored Envelopes. Checks take `O(B + S)` with a set for sensor IDs. Indexed inserts take `O(S log R + log E)`. Memory takes `O(B + S)`. Disk sync adds storage latency; this formula does not predict elapsed time.

Planning size	Estimated work and effect
Initial farm	10 devices, one sample per minute: 14,400 Readings/day and 432,000 in 30 days. Average intake is 0.167 envelopes/second.
100 times that load	1,000 devices: 1,440,000 Readings/day and 43,200,000 in 30 days. Average intake is 16.67 envelopes/second.
Index and disk estimate	`log2(R)` rises from about 19 to 26. A planning allowance of 1 KB per sample gives about 432 MB or 43.2 GB.

These are sizing assumptions, not measured throughput or known farm inventory. SQL B-trees use page fanout, so log2 values only compare growth. WAL files, backups, and free space add to the disk allowance. The 100-times case needs a host storage check and a commit benchmark. It does not establish that one LoRa channel carries that load.

4. Why this algorithm. A transaction prevents partial envelopes. A primary key and full content comparison give persistent duplicate checks. An in-memory set loses history after restart. A separate lookup followed by an unlocked insert races with another writer. A receipt before commit loses data after a crash. Generated Protobuf bytes avoid a custom hash or canonical JSON scheme.

5. Edge cases. Reject empty samples, unknown encodings, empty sensor IDs, negative times, and repeated sensor IDs. Equal times remain separate messages; order queries by time, then message ID. Unknown collection time stays zero. An identical retry returns `DUPLICATE`. Changed payload under the same key returns a conflict. Unapproved remote identities never reach storage. Disk full, busy timeout, and a failed commit send no receipt. A crash before commit leaves no new envelope; a crash after commit permits a matching retry.

4. Build-versus-reuse decisions

Versions below come from the checked source tree, not a claim about the newest release. `Cargo.lock` pins Rust dependencies. `gradle/libs.versions.toml` pins Java dependencies. Project code uses the Pulse proprietary course license [P3]. The team checks upstream license files and documentation on October 6, 2026. This table covers the significant parts of C3 and C5.

Piece	Build or reuse; license	Maturity, performance, and fit
Delivery lifecycle	Build; Pulse course license	Project-specific retry and receipt rules. Bounded scans suit 60 pending messages. Library containers provide the basic operations.
Atomic storage and duplicate rules	Build; Pulse course license	The key, transaction order, and content conflict rule belong to Pulse. Use SQLite transactions instead of custom recovery.
Decoder registry and sensor checks	Build; Pulse course license	US-04 needs a small encoding-to-decoder map. Each decoder defines its fields and units.
Local RPC and framing	Build the contract; Pulse course license	I3/I6 need a shared small frame, explicit errors, and Rust/Java vectors. Reuse generated codecs for the body.
Async I/O and timers	Reuse Tokio 1.53.1; MIT [W2]	Maintained runtime with documented support policy. Use bounded channels and timers on OS hosts. ESP32 runtime fit remains a D2 task.
Protobuf codecs	Reuse prost 0.13.5, Apache-2.0; protobuf-java 4.32.1, BSD-3-Clause [W3, W4]	Established schema tools produce compact binary messages. Shared vectors check language agreement. Reuse compiler output; keep field numbers stable.
Signature and verification	Reuse ed25519-dalek 3.0.0; BSD-3-Clause [W5]	Established library family; version 3 matches the SDK. Sign exact bytes. Reuse the implementation; measure board cost in D2.
Network stack	Reuse reticulum-sdk 2.3.4; MIT [W6]	Small upstream project; direct LoRa support is experimental. Pin this version. It provides paths and encryption. Hardware and Python interoperation remain required checks.
Database engine and driver	Reuse SQLite, public domain; Xerial 3.49.1.0, Apache-2.0 and BSD-2-Clause notices [W7, W8]	Mature embedded database; maintained JDBC driver. One writer fits initial intake. Test FULL commits on the actual host disk.
Message IDs	Reuse uuid 1.26.0; MIT OR Apache-2.0 [W9]	Established UUID parser and generator. Version 4 avoids a shared ID service. Treat a collision with changed data as a conflict.
JSON decoding	Reuse Gson 2.13.1; Apache-2.0 [W10]	Maintenance mode is sufficient for JSON parsing. Use strict parsing and explicit field checks. Adding it to runtime is a D1 target; it is a test dependency today.
Time and scheduling	Reuse Rust `Instant`, Java `System.currentTimeMillis`, and library timers; Rust MIT/Apache-2.0, OpenJDK GPL-2.0 with Classpath Exception [W11, W12]	Standard APIs avoid a custom date parser. Monotonic time controls retries. Epoch milliseconds label readings. No textual date input is needed.

Retain each required license and notice with distributed dependencies. The Pulse license does not replace dependency licenses. D1 selects software already present in the project where possible. A small upstream project has less support than SQLite or Protobuf; the SDK version pin and interoperation checks address that risk.

5. API contract

Transport, session, and common types

I3 and I6 use persistent TCP on `127.0.0.1:4401` by default. Configuration can select another loopback port from 1 to 65,535. One frame contains a 4-byte unsigned big-endian body length and one Protobuf `RpcEnvelope`. Body length is 1 to 1,048,576 bytes. Check the length before allocating the body. A bad length closes the connection. These methods use protocol error codes, not HTTP status codes.

The common wrapper is `RpcEnvelope{protocol_version: uint32 = 1, request_id: uint64, op: Operation, body: oneof}`. A request uses a required ID from 1 to 2^64-1, unique among that connection's pending requests. A response echoes that ID. An event uses ID zero. `op` and `body` must agree. Proto3 defaults do not make an application-required value optional.

Type or field	Exact rule
`Name`	Required ASCII string, 1-64 characters, using letters, digits, _, or -. Applies to origin, sensor, service, and destination names. Service `node` is reserved. D1 uses service names, not arbitrary hashes.
`message_id`	Required canonical lowercase UUID v4 string, 36 characters. Reuse it unchanged on retries. Within one Node it identifies one immutable Publication.
`PulseEnvelope`	`{envelope_version:uint32, message_id:string, origin_id:string, collected_at_ms:int64, samples:list<SensorSample>}`. Version is 1. Origin is Name. Time is optional, 0 when unset; otherwise 1 to 2^63-1 UTC milliseconds.
`SensorSample`	`{sensor_id:string, encoding:string, payload:bytes}`. Sensor is Name and is unique within the envelope. Encoding is required, 1-128 ASCII characters, and names an installed decoder.
Samples and payload	Required list with 1-16 samples. Each payload contains 1-262,144 bytes. The total local frame must fit the frame limit. Remote publication also requires its complete signed wrapper to fit 383 bytes.
`AckMode`	`ACK_NONE=0`, `ACK_RECEIVED=1`, `ACK_STORED=2`. Optional; default is `ACK_NONE`. Reject other values. Farm telemetry explicitly requests `ACK_STORED`.
`Response`	Success: `{ok:true, error:absent, health:absent}`; HEALTH adds HealthStatus. Failure: `{ok:false, error:{code:ErrorCode,message:string}, health:absent}`. The message explains the cause; callers use the code.

I5 carries `TransportPacket{payload:bytes, sender_identity:bytes[16], signature:bytes[64]}`. The payload contains `SignedPayload{ack:AckMode, body:data(bytes) | status(StatusUpdate)}`. Data bytes encode one PulseEnvelope. The signature covers the exact payload bytes. The configured public identity contains 64 bytes, shown as 128 hex characters. The 16-byte hash refers to that identity; it is not the full public key. Reticulum owns the route and hop limit [P3].

The 383-byte cap comes from the pinned SDK's encrypted single-packet limit. It includes the whole Pulse wrapper. A 1 MiB local frame does not imply a 1 MiB radio packet. Oversize remote data returns `INVALID_MESSAGE`; D2 can define fragmentation if required.

Key operations on C3

Method	Inputs, required values, and limits	Output and errors
PUBLISH / I3	Required destination:Name, message:PulseEnvelope. Optional ack:AckMode=ACK_NONE, want_acceptance:bool=false. Sender origin must match the local configured origin.	Success returns Response if acceptance is true. Failure always returns Response. INVALID_MESSAGE: bad fields, conflicting pending ID, unsupported enum, or oversize wrapper. SERVICE_UNAVAILABLE: no configured provider. QUEUE_FULL: reliable capacity is full. UNAUTHORIZED: origin binding fails.
SUBSCRIBE / I6	Required service:Name; it must name a service allowed for this local component. Repeat registration on the same connection has no extra effect. Only one active Hub consumer owns a storage service.	Success returns Response. Deliveries then use Event. INVALID_MESSAGE: invalid or reserved service. SERVICE_UNAVAILABLE: another consumer owns it. UNAUTHORIZED: the component cannot host it.
ACK / I6	Required message_id:string under the UUID rule; required status:DeliveryStatus=STORED. Optional detail:string="", 0-256 UTF-8 bytes. The calling connection must own the service's pending storage delivery.	Success returns Response and causes a signed remote storage receipt. INVALID_MESSAGE: unknown waiter, non-STORED status, or bad fields. UNAUTHORIZED: another connection attempts the receipt. A lost remote receipt triggers a new delivery; its new waiter permits another ACK.
HEALTH / I3 or I6	Required empty HealthRequest{}; no parameters. Wrapper version and request ID rules still apply.	Success returns Response with HealthStatus. INVALID_MESSAGE: version or operation/body mismatch. A closed connection yields a client I/O error.

These validation and ownership rules are target D1 rules. Current code does not enforce every name limit, local origin binding, or ACK connection check. ACK currently accepts other statuses. Add these checks before claiming the D1 contract is complete [P3]. Loopback restricts network access; it does not authenticate every local process. The first deployment uses one trusted service account and protected configuration files.

Key method inside C5

ReadingStore.store(envelope:PulseEnvelope, authenticatedSender:bytes[16], receivedAtMs:int64) returns StoreResult = STORED | DUPLICATE. All arguments are required. The limits match the common types. receivedAtMs is 1 to 2^63-1 UTC milliseconds. A2 defines the checks and transaction order. The method generates comparison bytes from the parsed envelope with the pinned Java Protobuf codec.

Internal error	Cause and caller action
INVALID_ENVELOPE	A field, count, time, or duplicate sensor fails a rule. Log the error; send no receipt.
UNKNOWN_SENSOR / UNSUPPORTED_ENCODING / INVALID_PAYLOAD	A sensor is not configured, a decoder is absent, or its parser/check fails. Reject the whole envelope.
IDENTITY_MISMATCH / CONFLICTING_DUPLICATE	The identity binding fails, or the same key carries different bytes. Keep committed data; send no receipt.
STORAGE_ERROR	SQLite returns busy timeout, disk full, or another transaction error. Roll back; send no receipt. The origin retries until expiry.

The current Java method returns Outcome and throws SQLException. The D1 method adds the listed checks and internal errors. These errors stay inside C5; they do not add unassigned wire enum values.

Events, status binding, and error recovery

Output	Exact shape, units, and conditions
`DeliveryEvent` / I6	`{destination:string, message:PulseEnvelope, authenticated_sender:bytes}`. Destination is the local service hash. Sender is exactly 16 bytes for remote data or empty for local data. C5 accepts an empty sender only for a configured local origin.
`StatusUpdate` / I5	`{message_id:string, status:DeliveryStatus, detail:string}`. Detail has at most 256 UTF-8 bytes. A signed expected peer supplies it. Match the peer and pending service before completion.
`StatusEvent` / I3	`{message_id:string, status:DeliveryStatus, detail:string}`. `RECEIVED=1`, `STORED=2`, `EXPIRED=3`, `FAILED=4`. Zero is invalid. Status has no numeric unit.
`HealthStatus`	`{inventory_id:string, boot_id:string, uptime_seconds:uint64, component:string, queue_depth:uint32, dropped:uint64, last_contact_ms:int64}`. Inventory is Name; boot ID is UUID v4; component is node. Uptime counts seconds. Queue depth counts pending Deliveries, including waiting items. Dropped counts best-effort removals since boot. Contact is epoch milliseconds, zero before any verified peer contact.

For local deliveries, C5 resolves an empty sender to the configured local identity before storage. The schema always stores a 16-byte identity hash. Collection and receipt times remain distinct. Status events omit destination in v1. Therefore, the Edge assigns distinct IDs to different destinations when one peer provides several services. A later schema can add explicit destination binding.

Error code or failure	When it occurs	Caller response
`INVALID_MESSAGE=1`	A supported frame has invalid Protobuf, version, operation/body, or application fields.	Fix the request. Do not retry unchanged invalid data. A malformed body gets request ID zero if no ID can be read.
`SERVICE_UNAVAILABLE=2`	A destination lacks a provider or the storage service has another active owner.	Check the service configuration and retry after the service becomes available.
`QUEUE_FULL=3`	Pending reliable Deliveries reach capacity.	Keep the sample at the caller and retry admission later. The Node does not own a rejected sample.
`UNAUTHORIZED=4`	A readable local request fails an origin, service, or ACK ownership check.	Fix the configuration or caller. Do not retry unchanged.
Invalid remote sender	Peer is unknown, signature is bad, or origin/service binding fails.	Receiver drops and logs the packet. It sends no trusted negative receipt. This implements AC-01.3 without replying to an untrusted sender.
Invalid length or closed stream	Zero/oversize frame, truncated stream, or I/O failure.	Reconnect after 100 ms; double the delay to a 5,000 ms cap. Re-subscribe. No event replay exists on the connection.

A valid `STORED` is the only success that proves Hub storage for `ACK_STORED`. Expiry means storage is unconfirmed; it does not prove that the Hub has no copy. A missing receipt after commit can produce this result. The Node reports the result so the technician can investigate.

Versioning. `protocol_version` versions local RPC; `envelope_version` versions data; both start at 1. Append optional Protobuf fields without reusing field numbers. Changing a field type, required rule, receipt meaning, or supported identity width breaks the contract. Such a change requires a version increase and coordinated peer upgrade. D0's identity-width change needs no migration of deployed farm devices because D0 is a design baseline. Future deployed v1 changes require a migration. A new sensor with an installed decoder does not require a new envelope version.

Example request and response

The example shows Protobuf JSON notation for readability. The wire uses Protobuf bytes with the length prefix. Bytes use base64. JSON uses strings for 64-bit integer fields. The sample payload decodes to `{"moisture":0.42}`. Moisture uses a fraction, not a percentage.

```
{
  "protocolVersion": 1,
  "requestId": "17",
  "op": "PUBLISH",
  "publish": {
    "destination": "hub-main",
    "message": {
      "envelopeVersion": 1,
      "messageId": "11111111-2222-4333-8444-555555555555",
      "originId": "field-01",
      "collectedAtMs": "1788912000000",
      "samples": [{
        "sensorId": "soil-1",
        "encoding": "application/json",
        "payload": "eyJtb2lzdHVyZSI6MC40Mn0="
      }]
    },
    "ack": "ACK_STORED",
    "wantAcceptance": true
  }
}
```

```
{
  "protocolVersion": 1,
  "requestId": "17",
  "op": "RESPONSE",
  "response": { "ok": true }
}
```

After the Hub commits, the field Node sends this event to the Edge.

```
{
  "protocolVersion": 1,
  "requestId": "0",
  "op": "EVENT",
  "event": { "status": {
    "messageId": "11111111-2222-4333-8444-555555555555",
    "status": "STORED",
    "detail": ""
  } }
}
```

An admission failure instead returns `response:{ok:false,error:{code:QUEUE_FULL,message:"reliable outbound queue full"}}` under request ID 17. Acceptance and `STORED` remain separate results.

6. Technology choices with justification

Database: SQLite through Xerial JDBC

Team skill fit: Yatharth uses SQL and PostgreSQL in his recorded work; Sujal uses SQL and database checks [P4]. Jacob also records Java experience. **Licensing:** SQLite uses public-domain terms; Xerial supplies Apache-2.0 and BSD notices [W7, W8]. **Community support:** SQLite has detailed transaction documentation, and Xerial maintains releases for SQLite changes. **Performance:** One writer, primary-key lookups, and WAL suit the assumed 0.167 envelopes/second; benchmark FULL commits before accepting the 100-times case. **Cost and hosting:** The database runs on the local Hub without a server subscription. We pass over PostgreSQL because a separate database service adds setup for the initial farm. Keep the option for measured writer or storage limits.

Backend: Rust Node with Tokio; Java 21 Hub

Team skill fit: Jacob records production Rust, C++, Java, and Linux work. Yatharth records Python, TCP/IP, and SQL work; Java Hub ownership adds a learning task [P4]. **Licensing:** Rust uses MIT/Apache-2.0, Tokio uses MIT, and OpenJDK uses GPL-2.0 with the Classpath Exception [W2, W11, W12]. **Community support:** Rust, Tokio, and OpenJDK have public documentation and maintained toolchains. **Performance:** Rust suits bounded packet state; a small Java consumer can validate and commit the initial intake. The JVM stays on the Hub, not the ESP32. **Cost and hosting:** Both programs run on existing Linux hosts without a paid backend service. We pass over a Spring service because C5 needs a local consumer, not a web server. We also pass over a Python Node to preserve Jacob's Rust work and the shared Rust libraries.

Front end: command-line tools and logs for D1

Team skill fit: Jacob records Linux and shell skills. Sujal and Yatharth record React experience if a later dashboard needs a web interface [P4]. **Licensing:** Pulse tools use the course license; the selected Rust/Java libraries keep their own licenses. **Community support:** Standard terminal tools and logging APIs have established documentation. **Performance:** Text output adds little work to the delivery path and needs no browser on field hardware. **Cost and hosting:** The technician uses the local host; no web hosting cost applies. We defer a React dashboard because US-01 to US-05 require delivery, configuration, and trust checks rather than a screen. D2 can define a separate dashboard without placing it inside the transport layer.

Messaging and hardware: Protobuf, Reticulum, and ESP32-S3 target

Team skill fit: Jacob's Rust and networking work supports the Node; Yatharth's industrial TCP/IP work supports bench tests. Sujal owns the sensor and hardware checks [P1, P4]. **Licensing:** prost uses Apache-2.0, protobuf-java uses BSD-3-Clause, and reticulum-sdk uses MIT [W3-W6]. Hardware has a purchase cost, not a software license. **Community support:** Protobuf has broad language support; the SDK has a smaller upstream group and experimental direct LoRa support. **Performance:** Compact bytes and a 383-byte wrapper cap suit low-rate messages; radio settings, RAM, and energy need board measurements. **Cost and hosting:** Reuse the owned ESP32-S3 boards and the LoRa board. Test Reticulum over TCP first. We pass over MQTT with a central IP broker because it adds an IP service to the field path. ESP32 firmware and any RNode fallback remain D2 choices, not proven D1 results.

Hosting: local Linux host with systemd

Team skill fit: Jacob records Linux deployment experience; Yatharth records database and industrial host work [P4]. **Licensing:** Linux uses GPL-2.0, and systemd uses LGPL-2.1-or-later for its main code [W13, W14]. Pulse runs as separate programs under these host tools. **Community support:** Linux and systemd have established operational documentation. **Performance:** Independent processes restart separately; local SQLite avoids network delay during commits. Restart still loses the current Node's in-memory queue. **Cost and hosting:** Reuse an existing laptop or Linux host for the prototype, with no cloud fee. A farm host needs reliable power and disk space. We pass over cloud-only hosting because farm internet loss must not stop local collection. The course constraints state $200; an advisor record states $1,000 [P5, P6]. D1 uses the smaller allowance and requires no new purchase.

Requirement checks and remaining implementation work

These checks define acceptance work. This document does not claim that the target changes pass them today.

Requirement or interface	Design location	Acceptance check
US-01; AC-01.1; I3-I6	A1	Disconnect the link for 10 minutes. Restore it before expiry. Check that a post-150-second retry delivers the same message ID.
US-02; AC-01.1; I7	Figure 1 and A2	Stop the Hub before commit, then restart it. Check that no partial envelope exists and the sender retains its copy.
US-02; AC-01.2	A2	Deliver the same envelope three times. Check one Envelope, one Reading per sensor, and a STORED receipt for each valid retry.
US-02; I3 and I5	A1 and API rules	Use two destinations with distinct expected peers. One STORED receipt removes only its Delivery. Check capacity across queued and waiting items.
US-04; I6 and I7	Sensor, Reading, decoder registry	Add an installed decoder and configured sensor. Reject unknown encodings, invalid JSON, and repeated sensor IDs without a partial commit.
US-05; AC-01.3	Origin binding and API errors	Send from an unknown peer and from a trusted peer with the wrong origin. Check zero Hub rows. A wrong peer's receipt removes no Delivery.
I6; receipt ownership	ACK contract	Call ACK from a different local connection. Check UNAUTHORIZED and no storage receipt.
D1 conflict rule; I7	A2	Change payload bytes under a stored key. Check CONFLICTING_DUPLICATE, unchanged rows, and no receipt.

Implementation order: Apply the SQL schema and full-envelope transaction first. Add strict decoder checks. Fix the queue key and total capacity count. Replace the five-retry cutoff with deadline-based retries. Add ACK ownership and request validation. Run the acceptance checks above. D2 then addresses persistent queue storage, radio integration, and the deferred components.

Project sources

- **P1:** `documents/sd1/assignments/assignment-5/D0_High_Level_Design.md`, with its original figure sources. D0 copies sit beside this file for comparison.

- **P2:** `User_Stories.md`: US-01 to US-05, UC-01, and AC-01.1 to AC-01.3.

- **P3:** The sibling `monorepo`, commit `c845c4e`, plus the inspected working files: `node/src/buffer.rs`, `node/src/node.rs`, `node/src/rpc.rs`, `hub/.../ReadingStore.java`, `HubMain.java`, `DecoderRegistry.java`, and `common/protocol/proto/pulse/v1/*.proto`. Dependency sources: `Cargo.toml`, `Cargo.lock`, `gradle/libs.versions.toml`, `MODULE.bazel`, and `LICENSE`.

- **P4:** The three biographies in `documents/sd1/assignments/assignment-1/`. These support the team experience statements.

- **P5:** `documents/sd1/assignments/assignment-3/Constraints_Essay.md`: $200 course budget and existing board inventory.

- **P6:** `docs/meetings/2026-09-11-first-advisor-meeting.md`: desktop-first plan, board tests, and the separate $1,000 allowance.

External sources and license checks

Links refer to primary project sources. Check date: October 6, 2026. Local crate manifests confirm the licenses of the pinned Rust versions.

ID	Source and supported decision
W1	SQLite WAL: commit sync and single-writer behavior.
W2	Tokio: runtime support and MIT license.
W3	prost: generated Rust codecs and Apache-2.0 license.
W4	Protobuf: schema tools; license.
W5	ed25519-dalek: signature implementation. Pinned 3.0.0 manifest states BSD-3-Clause.
W6	Reticulum SDK: MIT terms and experimental direct LoRa support. Pinned 2.3.4 packet source defines the 383-byte limit.
W7	SQLite copyright: public-domain terms.
W8	Xerial JDBC: maintenance, supported hosts, and license files.
W9	uuid: UUID generation and MIT/Apache-2.0 choice.
W10	Gson: maintenance mode and Apache-2.0 license.
W11	Rust licenses: MIT/Apache-2.0 terms.
W12	OpenJDK license: GPL-2.0 with the Classpath Exception.
W13	Linux license: GPL-2.0 license rule.
W14	systemd license: main-code LGPL-2.1-or-later rule and exceptions.

Submission files

The submission is `D1_Detailed_Design.pdf`. The folder also contains the Markdown text, SVG figure, PNG export, SQL schema, and Python export script. D0 copies remain unchanged. Each team member submits the same complete PDF separately. Course submission remains a team action.